

NERVOUS CONTROL

NERVOUS SYSTEM

- The nervous system is a means of coordinating activities and reactions to stimuli with the help of the following parts:
 1. **Receptors:** Sensory cells or organs that detect stimuli from the external and internal environment, transferring information (impulses) to the nerve cells.
 2. **Nerve cells (neurones):** specialised cells that transmit information in the form of nerve impulses.
 3. **CNS (central nervous system):** Made of the brain and spinal cord. It is part of the NS (nervous system) that coordinates and processes the information.
 4. **Effector:** a structure, gland or organ that responds to the stimuli.
 5. **PNS (Peripheral nervous system):** Nerves (cranial and spinal).

NEURONES

- Nervous system → nervous tissues → nerve cells (neurones).
- Each neurone consists of a cell body with a nucleus. The cell body has short cytoplasmic extensions known as dendrons that transmit impulses to the cell body. Dendrons branch into smaller extensions known as the dendrites.
- Nerve impulses are carried away from the cell body via long single extensions known as axons.
- There are THREE types of neurones, each with a different function:
 1. **Sensory neurones:** Transmit impulses from receptors to the CNS.
 2. **Relay neurones (a.k.a intermediate or connector neurones):** transmit impulses from sensory neurones to motor neurones.
 3. **Motor neurones:** Transmit impulses from the CNS to effectors.

MOTOR NEURONE

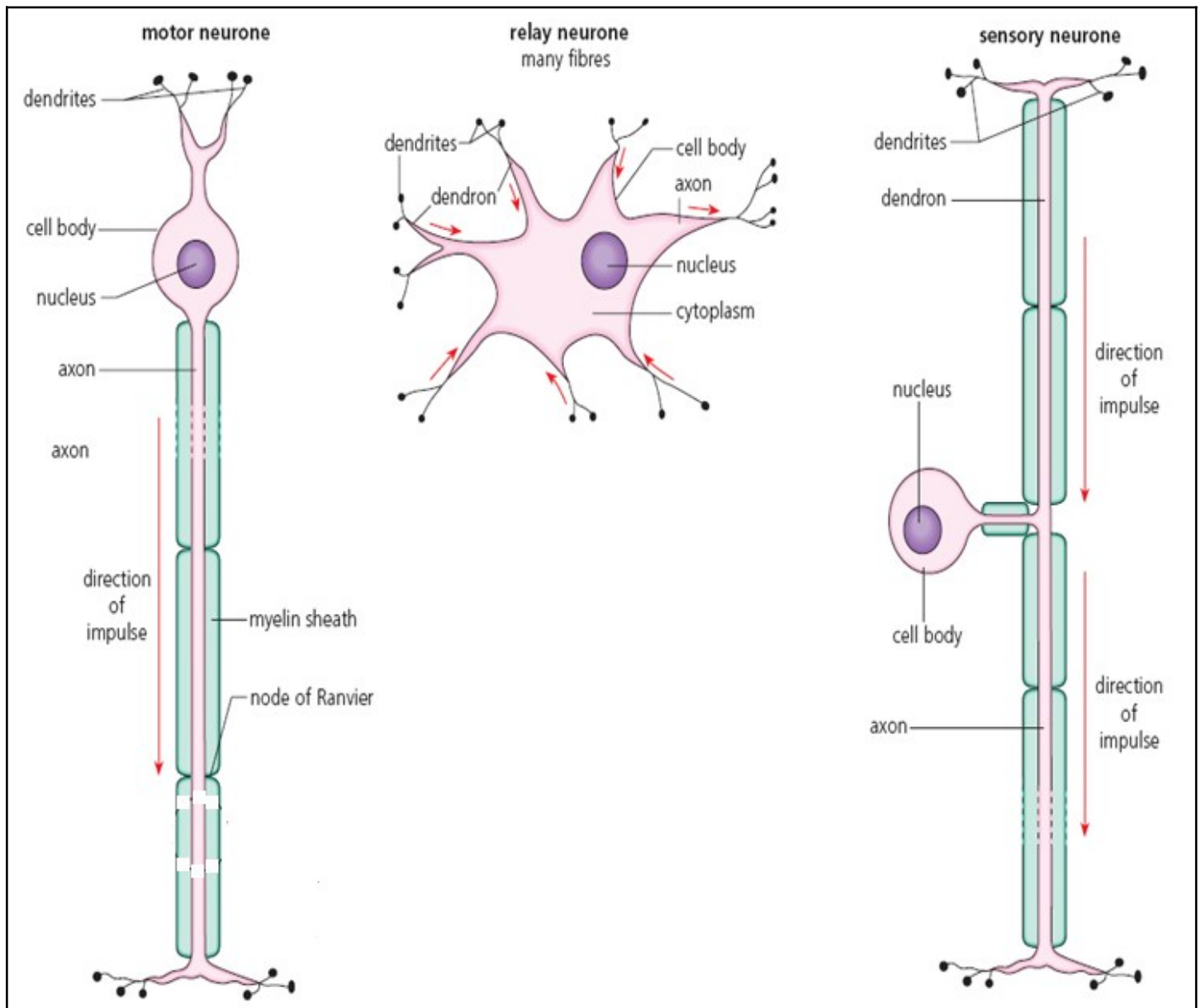
- Motor neurone's cell body lies within the spinal chord or brain.
- Dark specks that can be seen in the cytoplasm are regions of RER that synthesise proteins.
- A motor neurone has many branched dendrites through which it receives information.
- The axon of a motor neurone is much longer and conducts impulses over long distances. The cytoplasm of axon contains large number of mitochondria together with many vesicles containing chemical transmitters. These vesicles are involved in passing impulses to an effector.

SENSORY NEURONE

- Has a single dendron which brings impulses towards the cell body.
- It has a single axon which carries impulses away from the cell body.
- Similar basic structure as a motor neurone.

RELAY NEURONE

- A.k.a intermediate or connector neurone.
- Found entirely within the CNS.
- Have numerous short fibres. Each fibre is a thread like extension of a nerve cell.
- Relays impulses to other neurones.



DIAGRAMS: Three types of neurones of the nervous system

MYELIN

- Surrounding the neurones are different types of supporting cells called **glia** cells.
- **Glia** cells **nourish**, **protect** and **insulate** neurones.
- In the PNS, the axons of many neurones are enclosed by glia cells called Schwann cells.
- **Schwann cells** are wrapped around the axon many times in a spiral to form a thick lipid layer called the **myelin sheath**.
- The **myelin sheath** provides **physical protection**, **nourishment** and **electrical insulation** for the axon, which greatly speeds up the transmission of **impulses (action potentials)**.
- Between each pair of Schwann cells is a tiny uncovered junction or gap called the node of Ranvier.
- The myelin sheath and the junctions help increase the speed at which impulses are conducted.
- Not all neurones are myelinated.

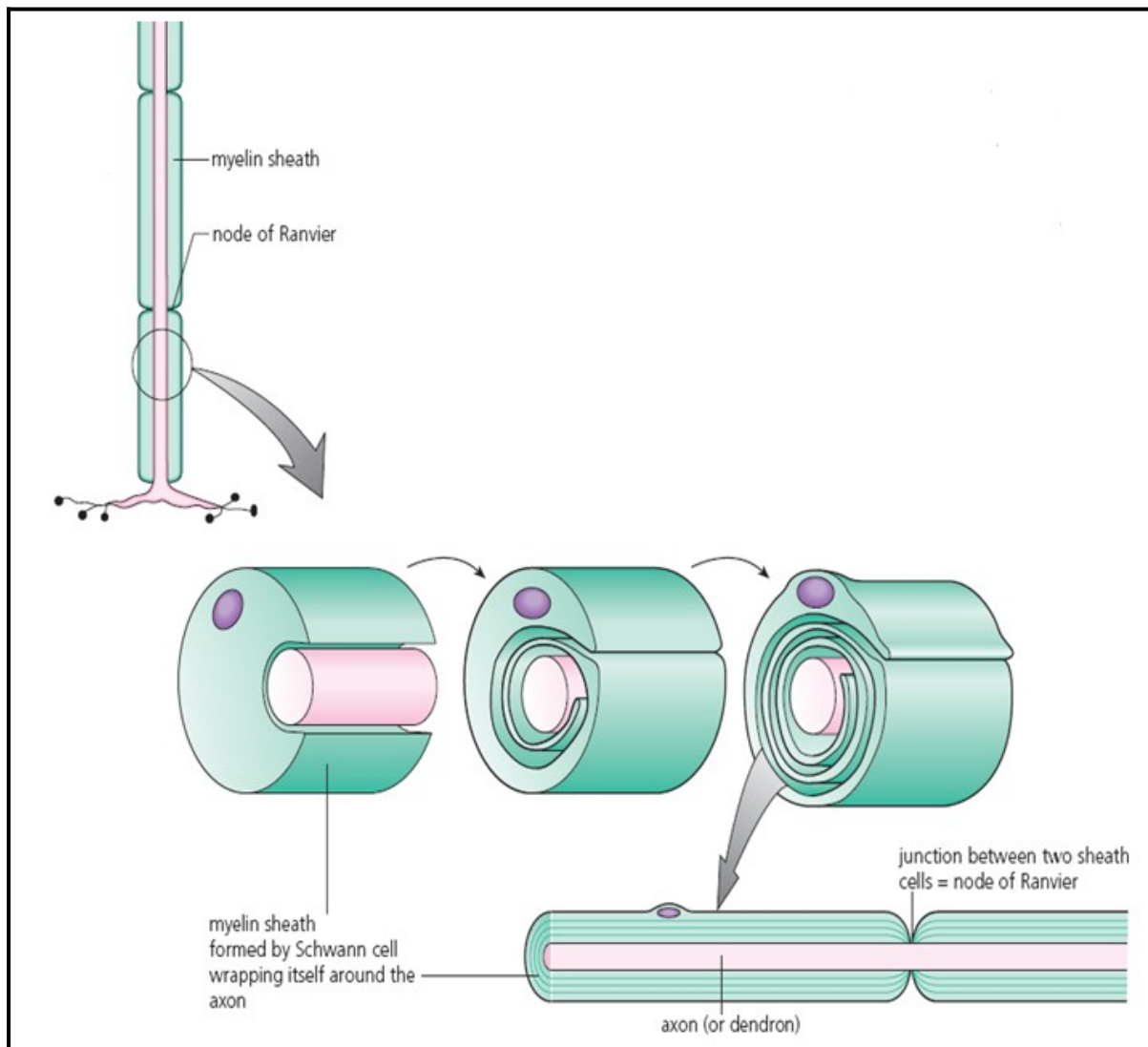
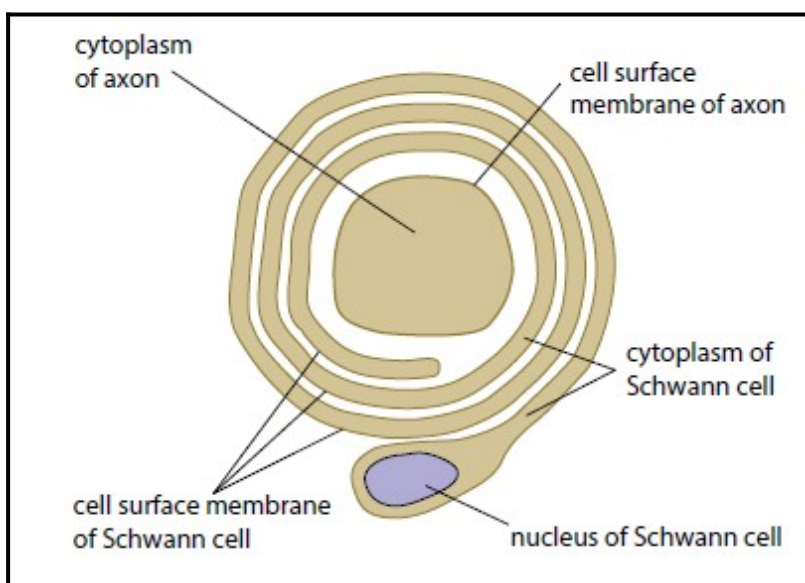


DIAGRAM: Most neurones have many companion cells called **Schwann cells**, which are wrapped around the axon many times in a spiral to form a thick lipid layer called the **myelin sheath**. The myelin sheath provides physical protection and electrical insulation for the axon, which greatly speeds up the transmission of action potentials. There are gaps in the sheath, called **nodes of Ranvier**. Not all neurones are myelinated.

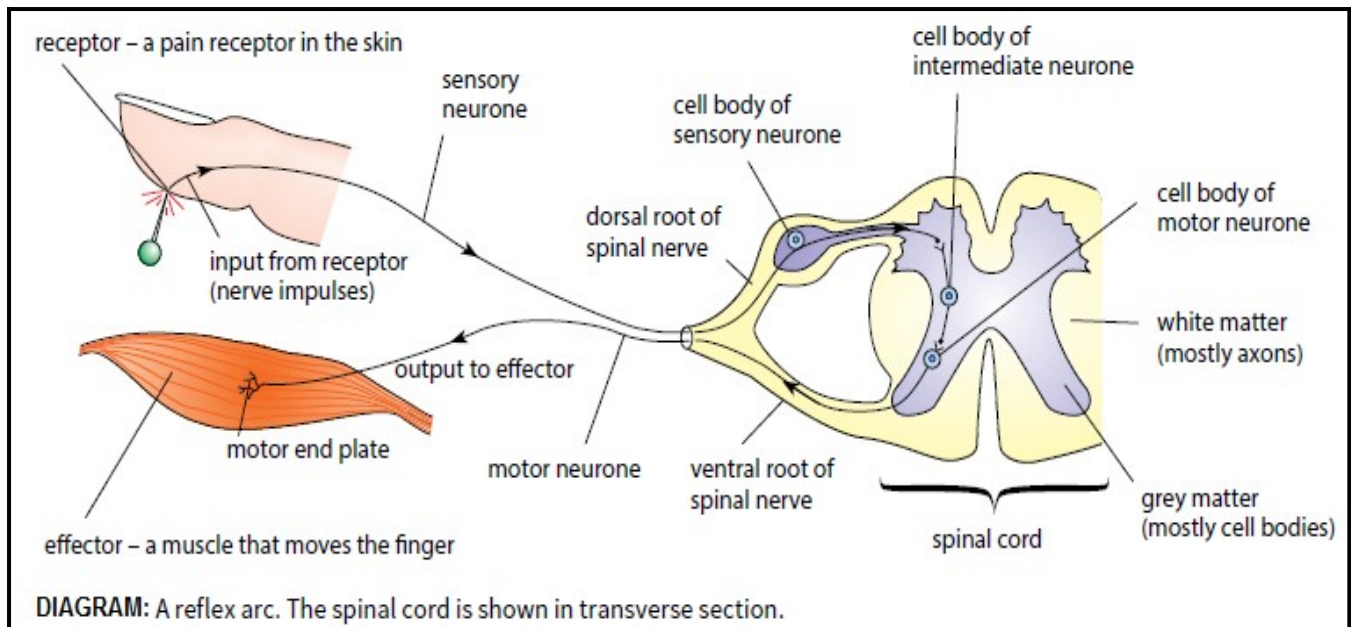


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DIAGRAM: TS: Myelin made of Schwann cell wrapped around axon

A REFELEX ARC

- A reflex arc is the pathway along which impulses are transmitted from a receptor to an effector without involving 'conscious' regions of the brain.
- Not all the reflex arcs may have intermediate neurones and the impulse may pass directly from the sensory neurone to motor neurone.



The diagram above shows the structure of a spinal reflex arc in which the impulse is passed from neurone to neurone inside the spinal cord.

REFLEX ACTION

- Within the spinal cord, the impulse will also be passed on to other neurones which take the impulse up the cord to the brain. This happens at the same time as impulses are travelling along the motor neurone to the effector.
- The effector therefore responds to the stimulus before there is any voluntary response involving the conscious regions of the brain.
- This type of reaction to a stimulus is called a **reflex action**. It is a fast, automatic response to a stimulus; the response to each specific stimulus is always the same.
- Reflex actions are a very useful way of responding to danger signals such as the touch of a very hot object on your skin or the sight of an object flying towards you.

QUESTION

Think of a reflex action other than the four already mentioned. State the precise stimulus, name the receptor that first detects this stimulus and the effector that responds to it, and describe the way in which the effector responds.

TRANSMISSION OF NERVE IMPULSES

- Neurones transmit electric impulses which travel very rapidly along the cell surface membrane.
- These impulses are not electric current but rather signals caused by very brief changes in the distribution of electric charge across the cell membrane. These are caused by very rapid movement of K^+ and Na^+ ions across the axon.

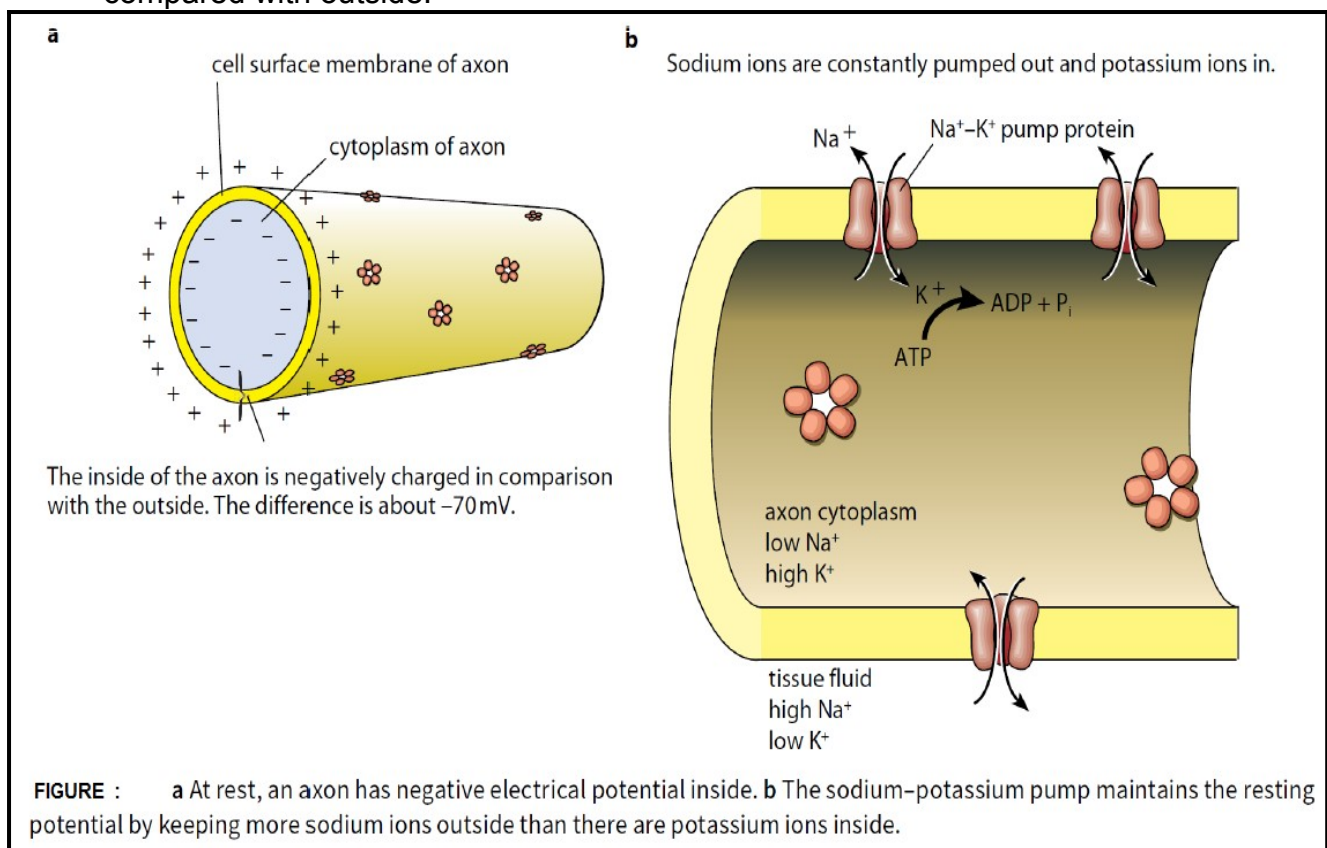
RESTING POTENTIAL

What is it?

- It is found that in a resting axon the inside is always slightly more negative than the outside.
- This difference in electric potentials is often between -60mV and -70mV and is called the resting potential.

How it is produced

- Produced and maintained by the $\text{Na}^+ - \text{K}^+$ pumps in the cell membrane.
- These pumps are membrane proteins that use energy from the hydrolysis of ATP to move these ions against their concentration gradient.
- Three Na^+ ions are removed for every two K^+ ions brought in.
- These protein channels are opened at all times and there are far more of these for potassium (K^+) than for sodium (Na^+).
- Therefore K^+ diffuses back out again much faster than Na^+ diffuses back in.
- There are also many large negatively charged molecules inside the cell that attract K^+ ions, reducing the chance that they will go out.
- As a result there is an overall build up of negative ions inside the membrane compared with outside.



ACTION POTENTIAL

- Action potential is caused by changes in the permeability of the cell membrane to Na^+ and K^+ ions.

Step #1: Depolarisation

- As well as the channels that are open at all times, there are some other channels that allow Na^+ and K^+ to pass through. These channels are called voltage-gated

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channels. They open and close when the potential deviates from the resting potential (i.e. they are closed at resting potential).

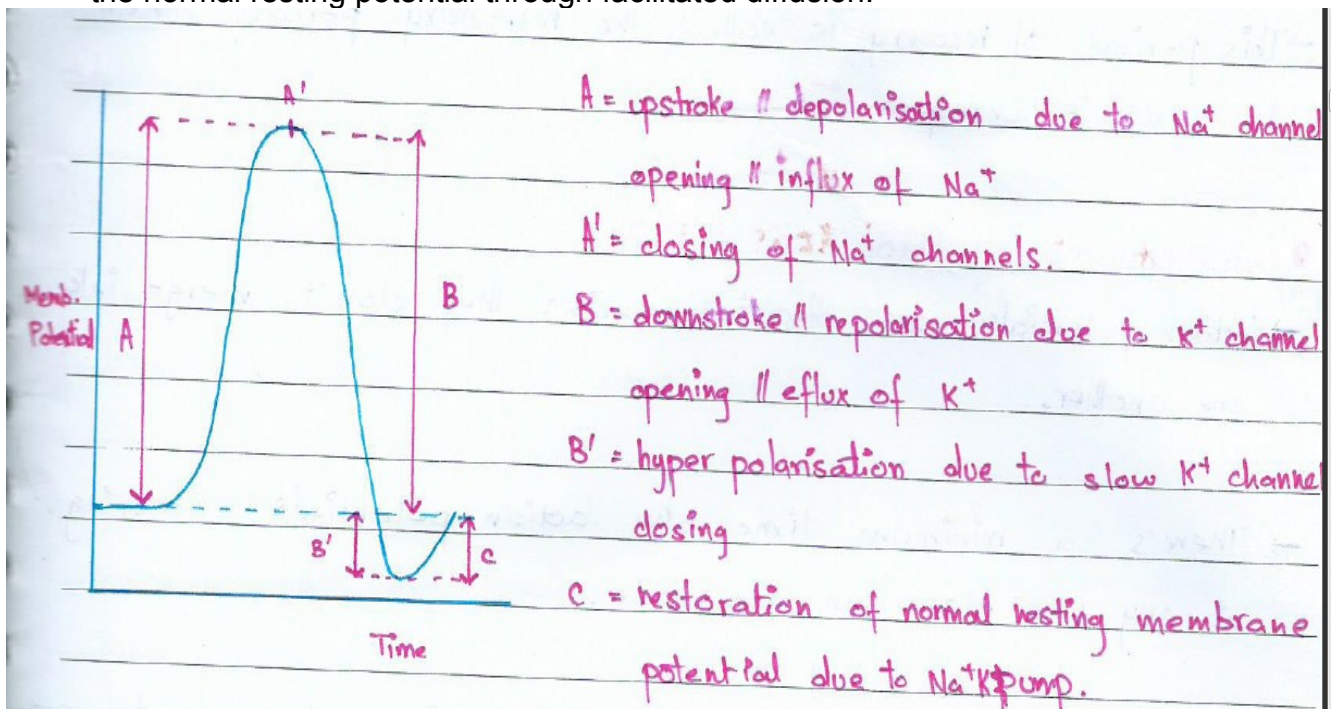
- The electric current used to stimulate the axon causes the opening of some voltage-gated channels, enabling the influx of Na^+ ions down their concentration gradient into the axon.
- The potential difference (p.d.) becomes less negative on the inside and this depolarisation triggers some more channels to open enabling more Na^+ ions in.
- If the p.d. reaches -50mV more channels open and the p.d. reaches $+30\text{mV}$ on the inside compared with the outside.
- Action potential only reaches if the p.d. reaches a value between -50mV and -60mV . This is the threshold potential and the action potential is not generated for a value less than this.

Step #2: Repolarisation

- After about 1ms , all the Na^+ voltage-gated channels close.
- The K^+ ion channels open and K^+ ions diffuse out of the axon down their concentration gradient.
- This removes the positive charge from the inside of the axon and returning to the normal resting potential of -70mV . This is called repolarisation.

Step #3: Hyperpolarisation

- The p.d. across the membrane briefly becomes more negative than -70mV .
- This is due to the slow closing of K^+ channels. K^+ channels now eventually close.
- The Na^+/K^+ pump now becomes active which pumps Na^+ out and K^+ in to restore the normal resting potential through facilitated diffusion.



TRY THIS QUESTION

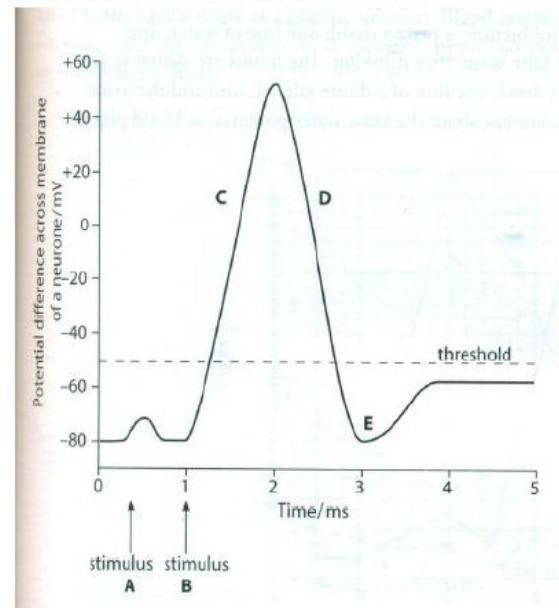
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5 The figure shows the changes in potential difference across the membrane of a neurone over a period of time. The membrane was stimulated at time A and time B with stimuli of different intensities.

a Stimulus B resulted in an action potential. Describe what is occurring at C, D and E. [6]

b Suggest why stimulus A did not result in an action potential being produced whereas stimulus B did. [2]

[Total: 8]



[Cambridge International AS and A Level Biology 9700104, Question 8, October/November 2007]

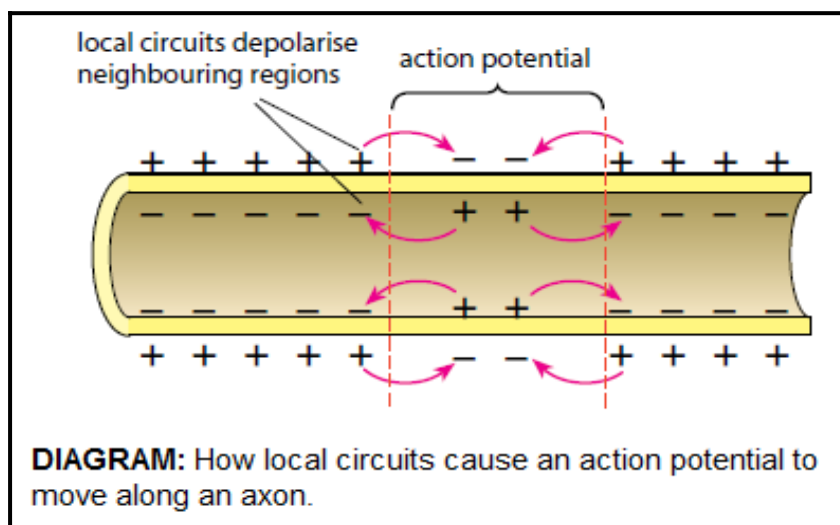
Expected answers

5 a

- C: depolarisation/the inside of the membrane becomes more positive/less negative; sodium ions/ Na^+ , flow in;
D: repolarisation/inside of the membrane becomes more negative/less positive; potassium ions/ K^+ , flow out;
E: hyperpolarisation/refractory period; more negative than resting potential; [6]

TRANSMISSION OF ACTION POTENTIAL

- In theory, an action potential at any point in an axon's cell membrane trigger the production of an action potential on either side of it.
- This happens because the temporary depolarisation of the membrane at the site of the action potential causes a **local circuit** to be set up between the depolarised region and the resting regions on the either side of it; hence causing action potentials in them as well.



- In practise, action potentials begin at one end and new action potentials are generated ahead and not behind.

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- This is because the region behind it will still be recovering from the action potential it has just had, and the sodium ion voltage-gated channels are 'shut tight' and cannot be stimulated to open, however great the stimulus. This period of recovery is the **refractory period** when the axon is unresponsive.

Consequences of the refractory period

There are several consequences of there being refractory periods.

- Action potentials are discrete events; they do not merge into one another.
- There is a minimum time between action potentials occurring at any one place on a neurone.
- The length of the refractory period determines the maximum frequency at which impulses are transmitted along neurones.

SPEED OF CONDUCTION

Speed of conduction is affected by:

1. Myelin sheaths

- Myelin speeds up the rate at which action potentials travel, by insulating the axon membrane.
- As Na^+ and K^+ ions cannot flow through the myelin sheath, so action potentials can only occur at the nodes of Ranvier, where all the channel proteins and pump proteins are concentrated.
- Hence the local circuits exist from one node to the next and therefore action potentials jump from one node to the next at a distance of 1 to 3mm.
- This is called **saltatory conduction** (from the Latin *saltare* meaning 'to leap') and this can increase the conduction speed by 50 times than that of an unmyelinated axon of the same diameter.

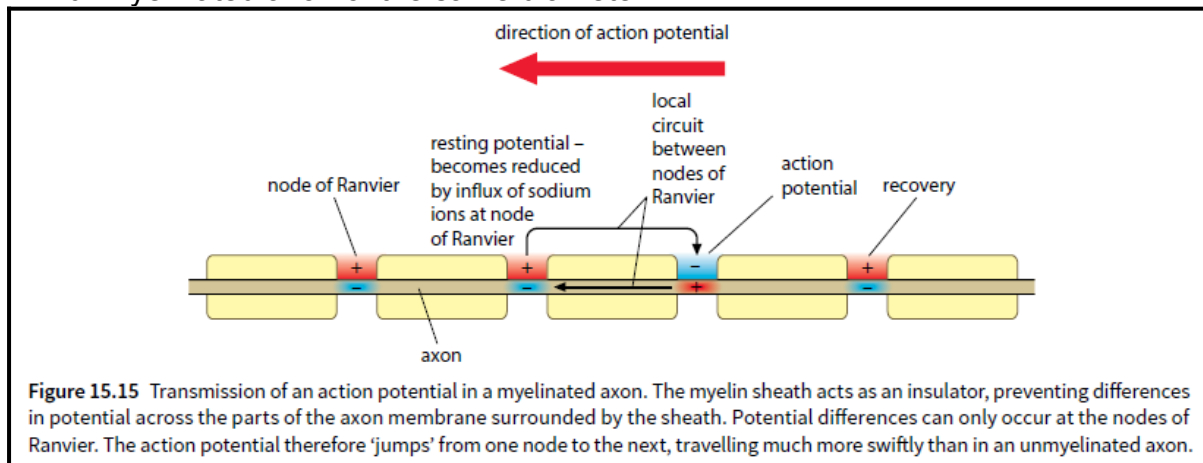


DIAGRAM: Saltatory conduction

2. Diameter

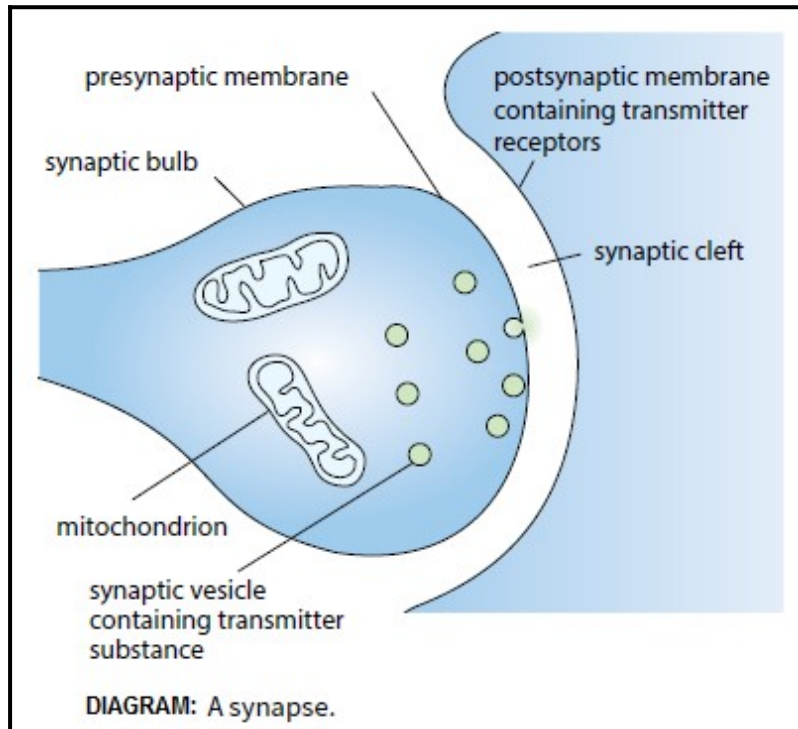
- The Greater the diameter of the axon, the greater the speed of conduction.
- This is because there is much less resistance in thicker axons than in thinner ones.

SYNAPSES

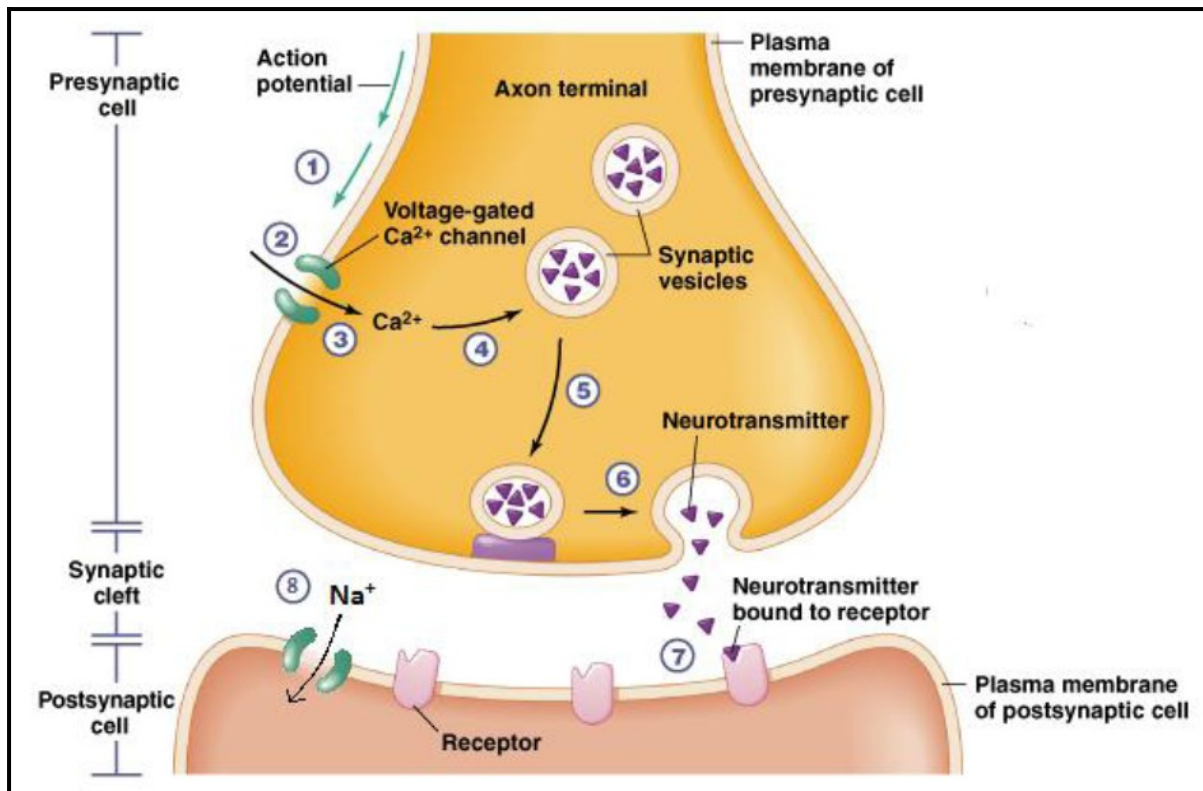
- When two neurones meet, they don't touch. Instead there is a very small gap about 20nm between them called the synaptic cleft.
- The parts of the two neurones near the cleft plus the cleft itself make up a synapse.

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- Synapse can be a junction between two neurones or between a motor neurone and a muscle cell.



MECHANISM OF SYNAPTIC TRANSMISSION



1. Action potential arrives at presynaptic neurone.
2. Stimulates opening of **voltage-gated channels** for Ca^{2+} ions
3. Ca^{2+} ions **diffuse** into cytoplasm of presynaptic membrane.
4. Ca^{2+} ions move to vesicles containing acetylcholine (ACh).
5. Ca^{2+} cause vesicles containing acetylcholine (ACh – a type of neurotransmitter) to move towards the presynaptic membrane.

6. Vesicle **fuses** with the membrane; Ach is released and **diffuses across** the synaptic cleft.
7. Ach **temporarily binds** to receptor proteins on the postsynaptic membrane; causes chemically-gated ion channels for Na^+ to open.
8. Na^+ ions diffuse through postsynaptic membrane and **depolarizes the membrane**. This generates an action potential.
9. **Recycling of Ach**: the enzyme acetylcholinesterase catalyses the hydrolysis of each Ach molecule into acetate and choline. This stops continuous production of action potentials.



10. The choline is taken back into the presynaptic neurone where it combines with acetyl CoA to form Ach.



11. The Ach is then transported into the presynaptic vesicles, ready for the next action potential.

THE FATE OF ACETYLCHOLINE (Ach)

- If Ach remained bound to the postsynaptic membrane, the Na^+ channels would remain open and the postsynaptic neurone would be permanently depolarised.
- The synaptic cleft contains an enzyme called acetylcholinesterase which catalyses the hydrolysis of each Ach molecule into acetate and choline.
- The choline is taken back into the presynaptic neurone where it combines with acetyl CoA to form Ach. The Ach is then transported into the presynaptic vesicles, ready for the next action potential.
- From the initial arrival of the action potential and the reformation of Ach, all the events take about 5 to 10 ms to happen.

ROLES OF SYNAPSES

1. Synapses ensure one way transmission.

- Impulses can only pass in one direction at synapses; because neurotransmitter (Ach) is released on one side and its receptors are on the other.
- Chemical transmission cannot occur in the opposite direction.

2. Synapses allow integration of impulses.

- Each sensory neurone has many branches at the end of its axon that form synapses with many relay neurones.
- The cell body of a motor neurone has many terminations from many relay neurones.
- Motor neurones only transmit impulses if the net effect of the relay neurone is above the threshold level. If the depolarisation of the postsynaptic membrane reaches the threshold level, only then an impulse is sent in that neurone.
- This disables the low frequency impulses from travelling in the sensory neurones to reach the brain and hence preventing the overloading of brain from sensory information.

3. Synapses allows interconnection of nerve pathways

- This happens in two ways;
 - (i) Individual sensory and relay neurones have axons that branch to form synapses with many different neurones. This means that information from one

neurone can spread out throughout the body to reach many relay neurones and many effectors, such as during dangerous situations.

- (II) There are many neurones that terminate on each relay and motor neurone as they have many dendrites to give a large surface area for many synapses; this allows one neurone to integrate the information from many different parts of the body – something that is essential for decision making in the brain.

4. Synapses are involved in memory and learning

- For example, if our brain frequently receive information about two things at the same time (e.g. the sound of a particular voice and the sight of a particular face) then it is thought that new synapses form in our brain that link the neurones involved in the passing of information along the particular pathway from your ears and eyes.